How We Refactor, and How We Know It

Emerson Murphy-Hill
Portland State
(now UBC)



Chris Parnin
Georgia Tech



Andrew Black
Portland State



Thanks to Ki-Yung Ahn, Barry Anderson, Lee Beckman, Brett Cannon, Tim Chevalier, Sara Dadizadeh, Sam Davis, Danny Dig, Rafael Fernandez-Moctezuma, Markus Keller, Rashawn Knapp, Andy Ko, Seonah Lee, Peng Li, Chuan-kai Lin, Ralph London, Gail Murphy, Immad Naseer, Fini Nitschke, Jinwgen Owen Ou, Sarah Rastakar, Claudia Rocha, Suresh Singh, Petcharat Viriyakattiyaporn, and Bryant York for assistance, as well as the National Science Foundation for partially funding this research under CCF-0520346.

SOME RIGHTS RESERVED

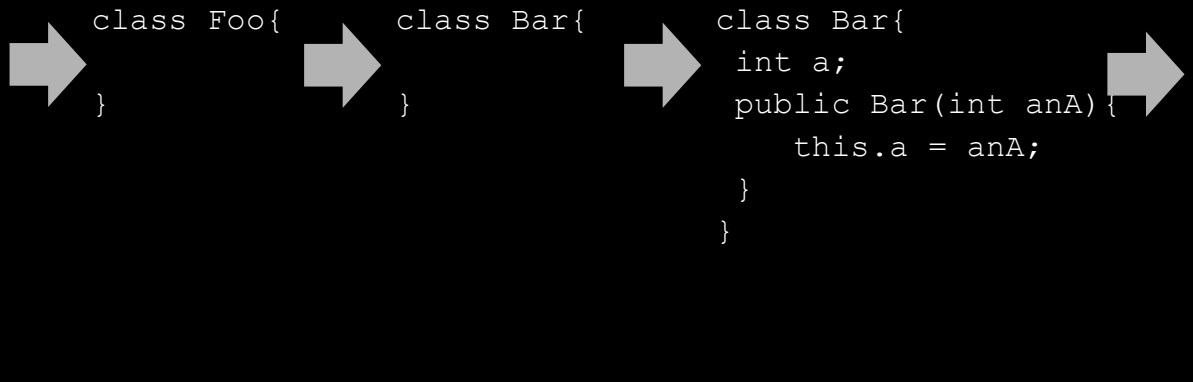








Introduction to Refactoring



Introduction to Refactoring

```
class Foo{
}
```

```
class Bar{
}
```

```
class Bar{
 int a;
 public Bar(int anA){
    this.a = anA;
 }
}
```

```
class Bar{
 int a;
 private Bar(int anA){
    this.a = anA;
 }
 public static Bar create(int anA){
    return new Bar(anA);
 }
}
```

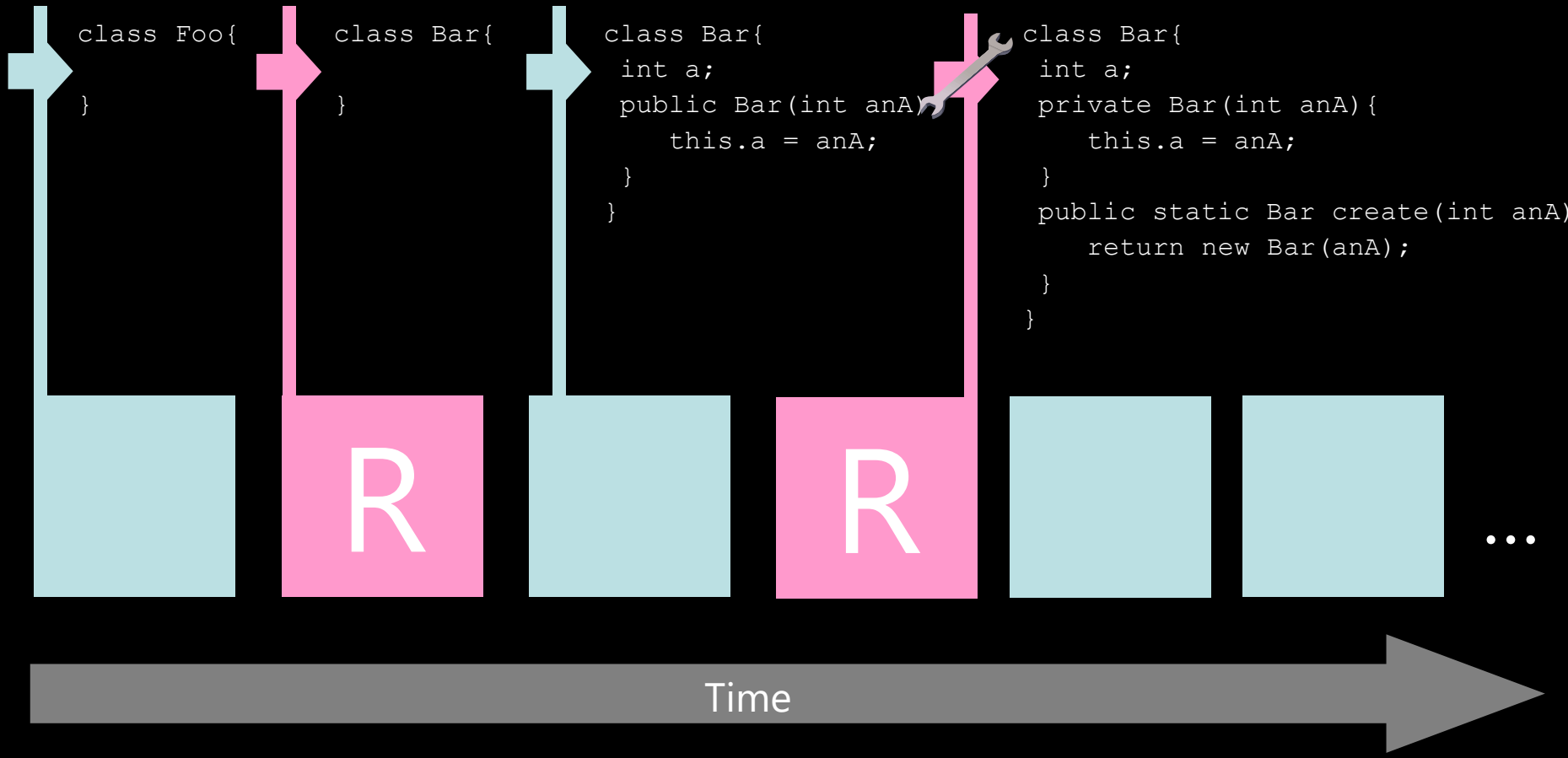
Correctness

&

Speed



Introduction to Refactoring



3 Research Questions

Do Programmers Usually Floss Refactor?



root-canal refactoring

Do Programmers Refactor Often?



Do Programmers Use Refactoring Tools Often?



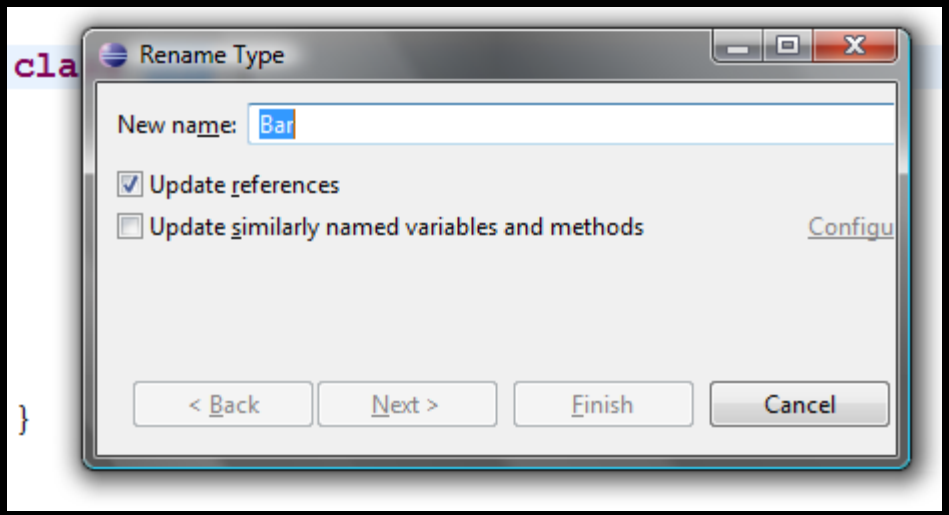
Do Programmers Usually Floss Refactor?

Do Programmers Usually Floss Refactor?

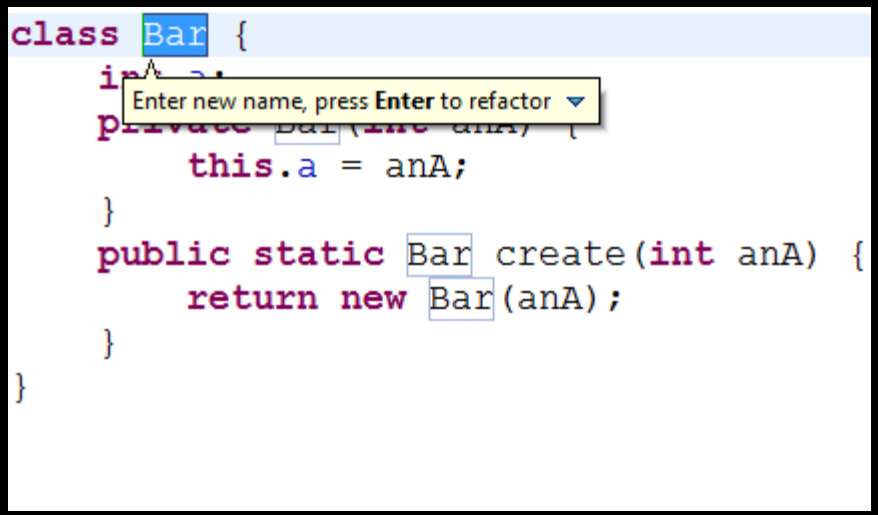
Importance

If floss refactoring is more common, then tools should support that strategy

A tool user interface optimized for "root canals"



A tool user interface optimized for "flossings"



Do Programmers Usually Floss Refactor?

What We Already Know

Floss refactoring better by prescription:

> In almost all cases, I'm opposed to setting aside time for refactoring. In my view refactoring is not an activity you set aside time to do. Refactoring is something you do all the time in little bursts. – *Martin Fowler*

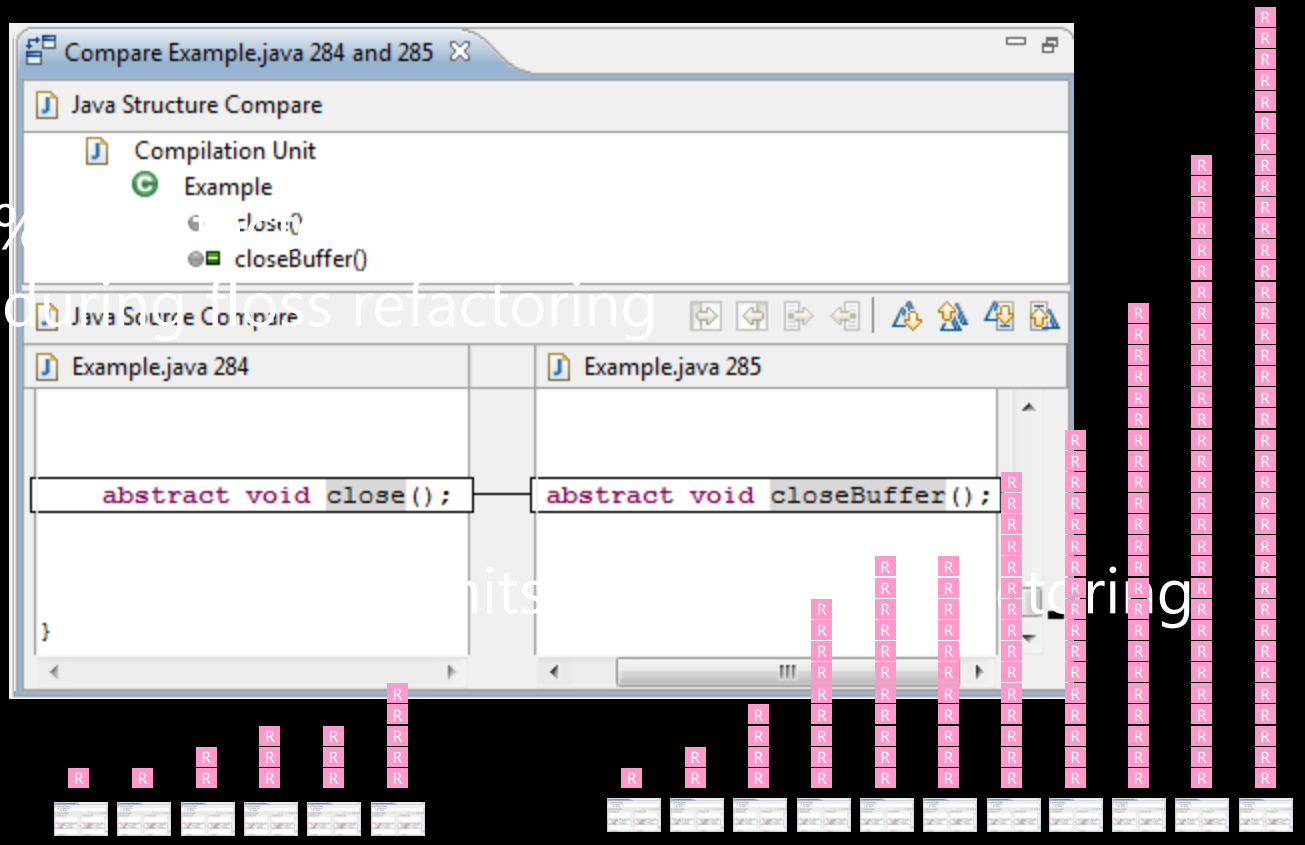
> Avoid the temptation to stop work and refactor for several weeks… Have your team get used to refactoring as part of their daily work. – *James Shore*

Case studies describe root canal refactorings:

- Pizka [2004]
- Bourqun and Keller [2007]

Do Programmers Usually Floss Refactor?

What We Did and Results



91%

during floss refactoring

its toring

Root Canal (Pure) Refactoring





Floss (Mixed) Refactoring

Do Programmers Usually Floss Refactor?

3 Research Questions

Do Programmers Usually Floss Refactor?



Yes

Do Programmers Refactor Often?



Do Programmers Use Refactoring Tools Often?



Do Programmers Refactor Often?

Do Programmers Refactor Often?

Importance

If programmers refactor often

- then refactoring research matters
- then refactoring tools may significantly speed up and improve the correctness of programming

Do Programmers Refactor Often?

What We Already Know

According to Xing and Stroulia [2006]:

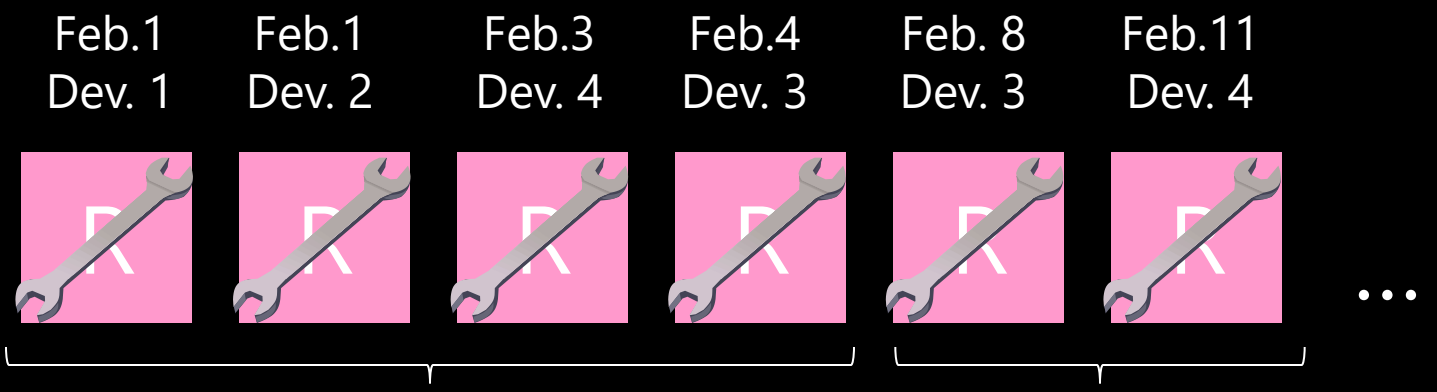
- Refactoring is frequent, in the Eclipse project
- But their use of an automated detection tool means the results are only a rough estimate

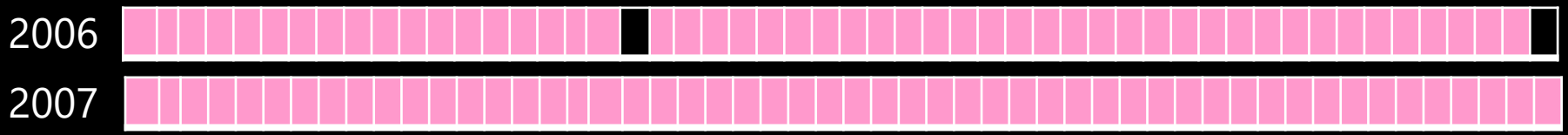
Do Programmers Refactor Often?

What We Did + Results

4 Developers on the Same Open Source Project

Refactoring Tool Logs in 2006/2007





But this is tool-based refactoring.
If many refactorings are done without tools,
then refactoring is even more frequent.

Do Programmers Refactor Often?

3 Research Questions

Do Programmers Usually Floss Refactor?

Yes



Do Programmers Refactor Often?



versus

Yes



Do Programmers Use Refactoring Tools Often?



versus



Do Programmers Use Refactoring Tools Often?

Do Programmers Use Refactoring Tools Often?

Importance

If programmers underuse refactoring tools:

- then there's potential for tool improvements
- then programmers are needlessly introducing errors or refactoring slowly

Do Programmers Use Refactoring Tools Often?

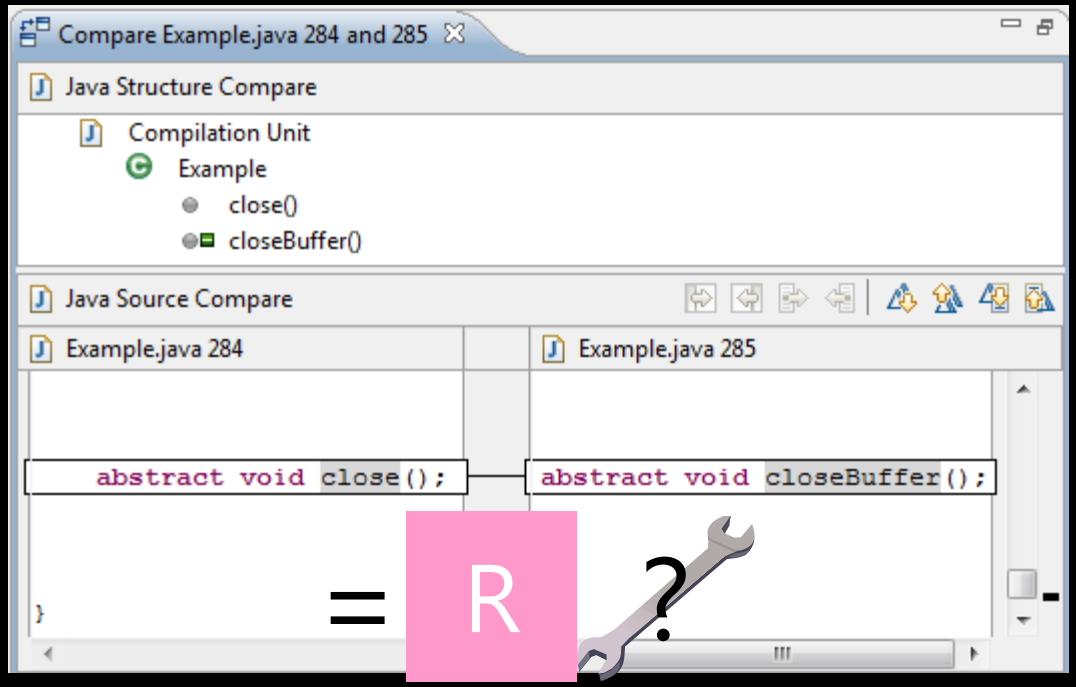
What We Already Know

From our previous work

- 2 in 16 programming students used them (and even then, 20% and 60% of the time)
- Agile developers (in 2007, n=112) estimate that they use them only 68% of the time
- 6 of 42 people who used Eclipse on networked PSU computers used refactoring tools

Do Programmers Use Refactoring Tools Often?

What We Did



Dev. 1 Commit
Feb. 1, 15:01

Dev. 1 Commit
Feb. 1, 15:23

Jan. 29, 08:40
Dev. 2

Feb. 1, 15:11
Dev. 1

Feb.1, 15:59
Dev. 1

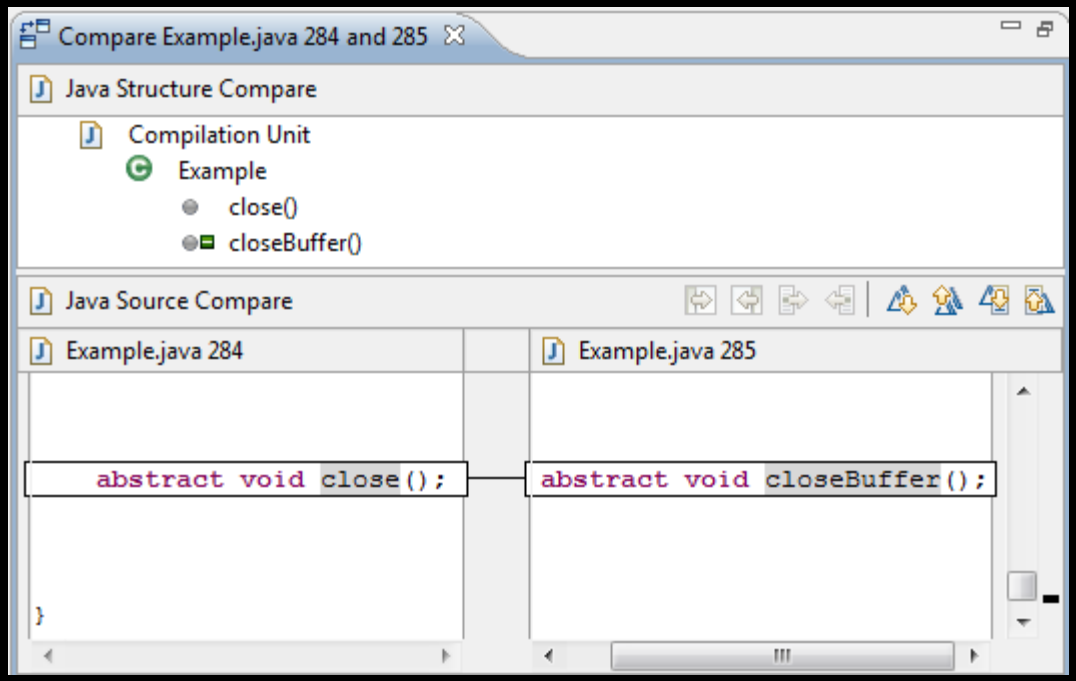
Feb.1, 16:00
Dev. 2

Feb.1, 12:23
Dev. 1

Do Programmers Use Refactoring Tools Often?



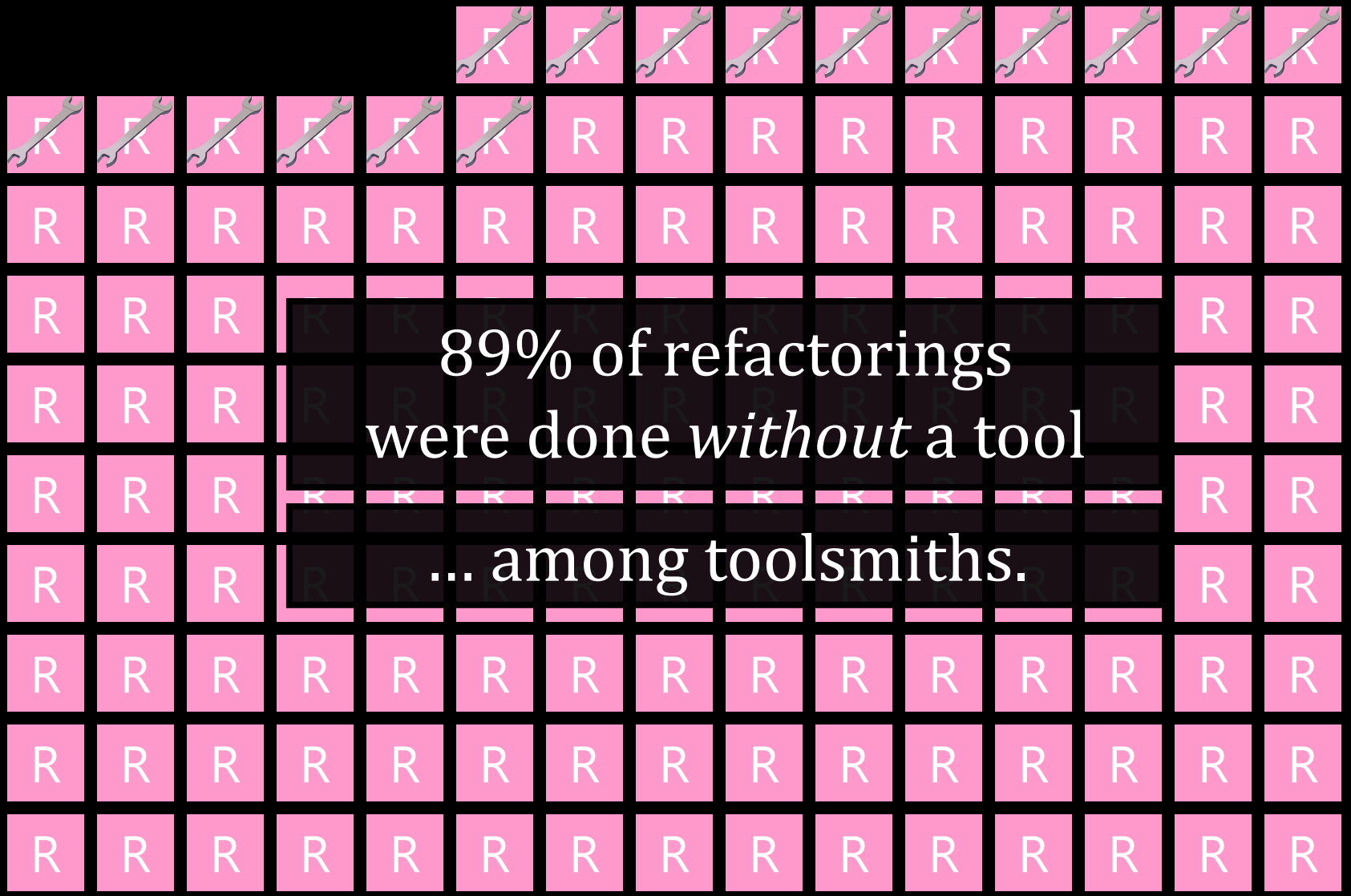
What We Did



Do Programmers Use Refactoring Tools Often?



The Results



Do Programmers Use Refactoring Tools Often?

3 Research Questions

Do Programmers Usually Floss Refactor?

Yes



Do Programmers Refactor Often?

Yes



Do Programmers Use Refactoring Tools Often?



versus **No**



6 More Findings

The kind of refactoring performed with tools differs from the kind performed manually

Toolsmiths use a wider array of refactoring tools than tool users

About 40% of tool-initiated refactorings occur in batches

About 90% of tool-initiated refactorings do not require configuration of the tool

Messages written by programmers in version histories are unreliable indicators of refactoring

About 40% of refactorings will not be detected by most mining tools that detect refactoring from version histories

Conclusion

The more ways we look at how people refactor, the more confidence we have that we understand how people refactor

In studying refactoring from new perspectives, our findings have confirmed – and refuted – previous knowledge about how people refactor